Causal LTI systems by difference / differential equation

An important class of discrete-time systems is that for which the input and output are related through linear constant-coefficient difference equation.

e.g. simple model for the balance in bank account from month to month. If 10% is the interest and $x[n]$ is the net deposit then balance at the end of the month is

$$y[n] = 1.01\, y[n-1] + x[n]$$

The general form of Nth-order linear constant-coefficient difference equation is —

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k]$$

Similarly, for continuous systems, systems where the input and output are related through a linear constant-coefficient differential equation.

The general form of such equations is —

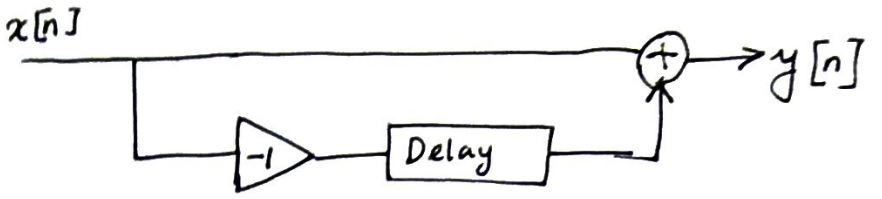
$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

ORDER — Highest derivative of output.

The solution of the equations

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{d t^k} = 0 \qquad \sum_{k=0}^{N} a_k y[n-k] = 0.$$

$R \rightarrow$ right shift

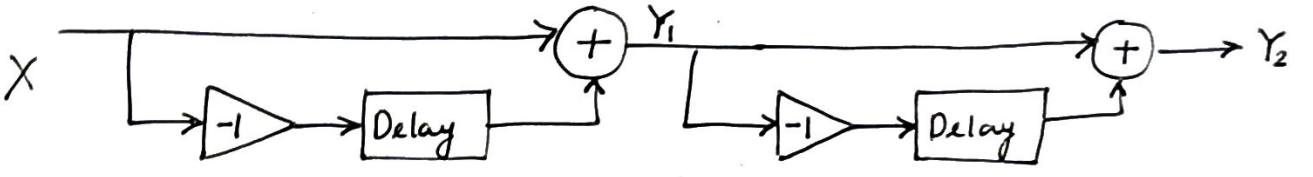


$$\boxed{Y = X - RX = (1-R)X}$$

Signal as a whole

$$\boxed{y[n] = x[n] - x[n-1]}$$

Difference equation



$$Y_1 = (1-R)X$$

$$Y_2 = (1-R)Y_1 = (1-R)(1-R)X$$

$$= (1 - 2R + R^2)X$$

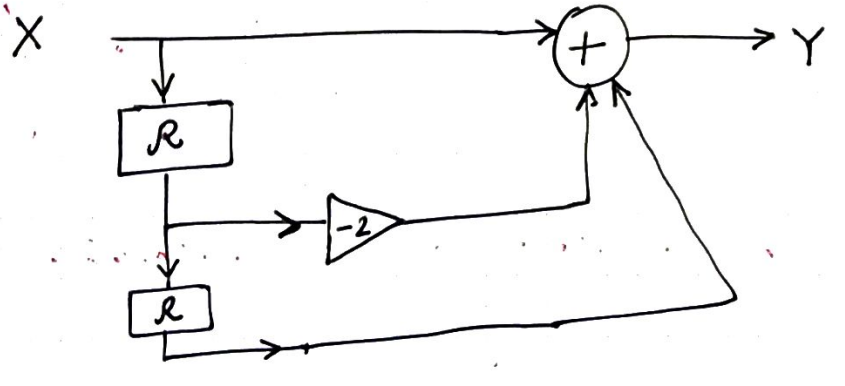
The equivalence is only true if they start at rest

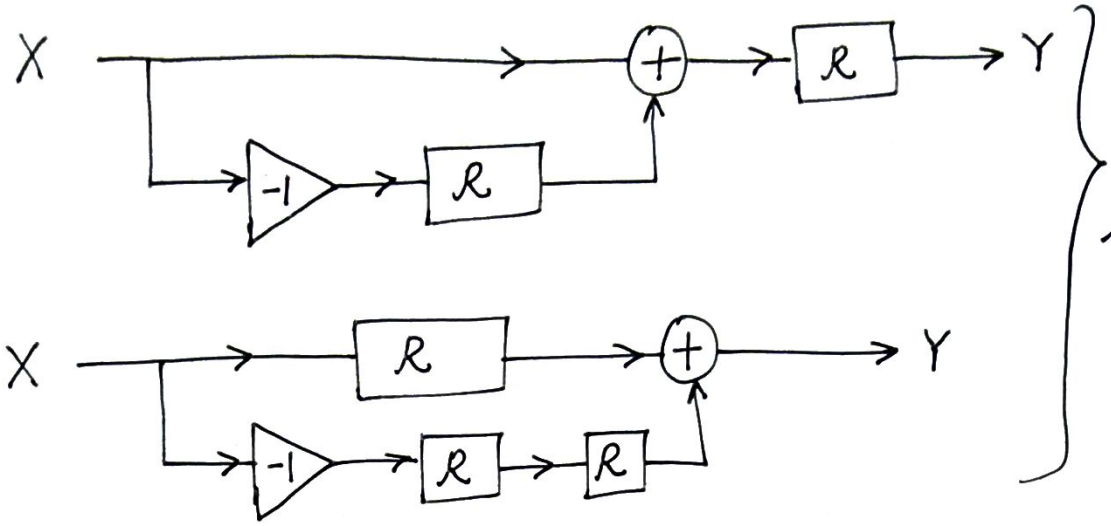
$$y_2[n] = y_1[n] - y_1[n-1]$$

$$= (x[n] - x[n-1]) - (x[n-1] - x[n-2])$$

(Delays may not be same)

$$y_2[n] = x[n] - 2x[n-1] + x[n-2]$$ ← equivalent diffence equation.





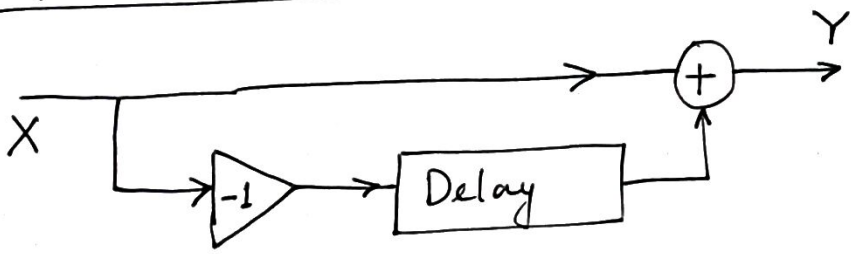
This are equivalent

$$Y = R(1-R)X$$

$$Y = RX - R^2X = (R-R^2)X$$

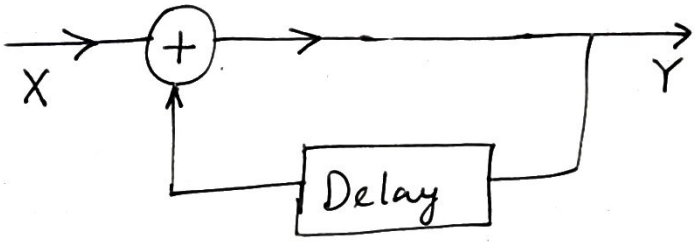
Distributive property

Explicit and Implicit Rules:



$$Y = (1-R)X$$

There is a recipe



$$Y = RY + X$$

$$\Rightarrow (1-R)Y = X$$

Recipe for finding i/p from o/p.

This does not tell how to get.

But how does one solve such a constraint?

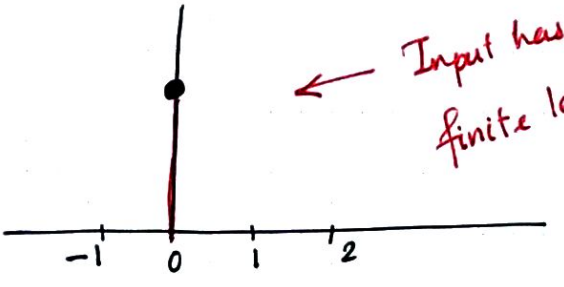
Accumulator:

$$x[n] = \delta[n] \qquad y[n] = x[n] + y[n-1]$$

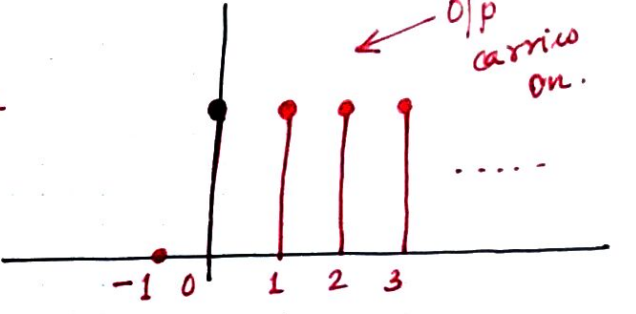
$$y[0] = x[0] + y[-1] = 1 + 0 = 1$$

$$y[1] = 0 + 1 = 1$$

$$y[2] = 0 + 1 = 1$$

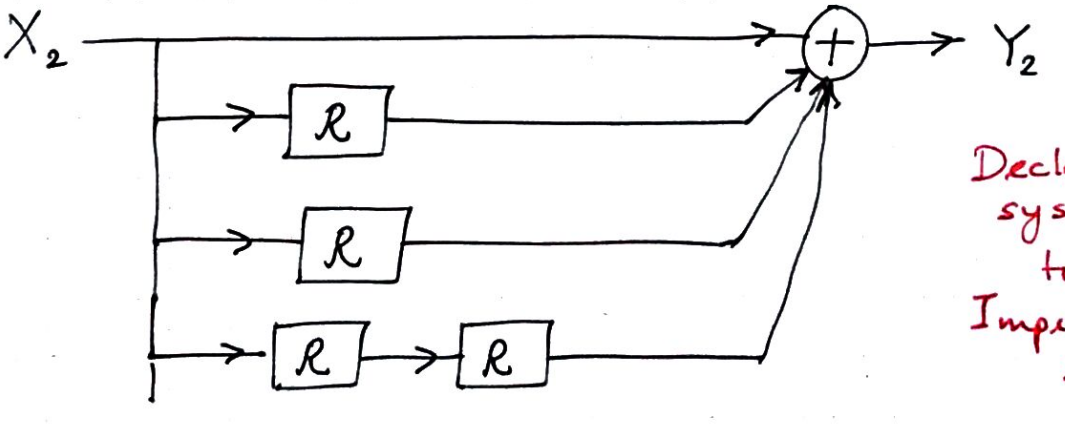


Input has finite length



o/p carries on.

Persitent response to a transient input.



Declarative system to Imperative system

$$(1-R)Y_1 = X_1 \overset{?}{\Longleftrightarrow} Y_2 = (1 + R + R^2 + R^3 + \cdots)X_2$$

Proof: Let $X_2 = X_1$

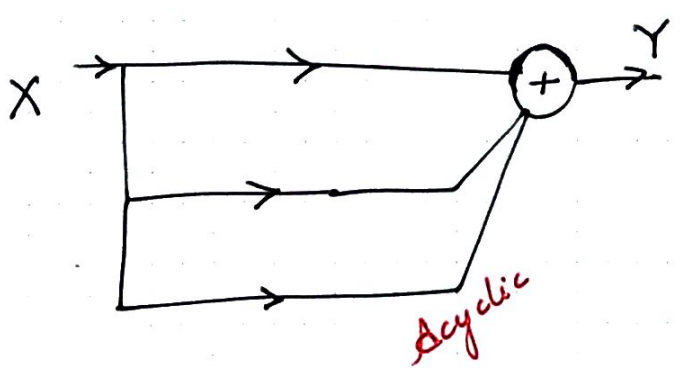
$$Y_2 = (1 + R + R^2 + R^3 + \cdots) X_2$$
$$= (1 + R + R^2 + R^3 + \cdots) X_1$$
$$= (1 + R + R^2 + R^3 + \cdots)(1 - R) Y_1$$
$$= (1 + R + R^2 + R^3 + \cdots) Y_1 - (R + R^2 + R^3 + \cdots) Y_1$$
$$= Y_1$$

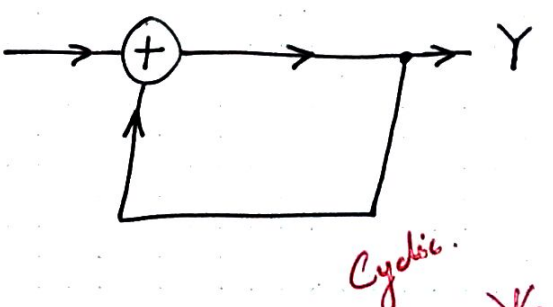
$\therefore$ $(1 + R + R^2 + R^3 + \cdots)$ and $(1 - R)$ are reciprocal.

$$(1 - R) Y = X \Rightarrow Y = \left(\frac{1}{1 - R}\right) X$$

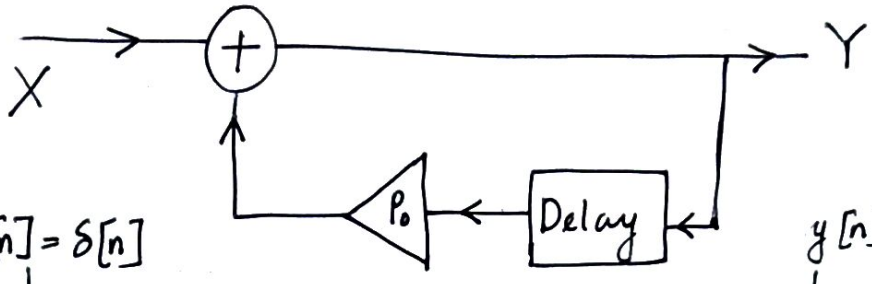
→ A transient input leads to a transient input for a system with imperative difference equation.

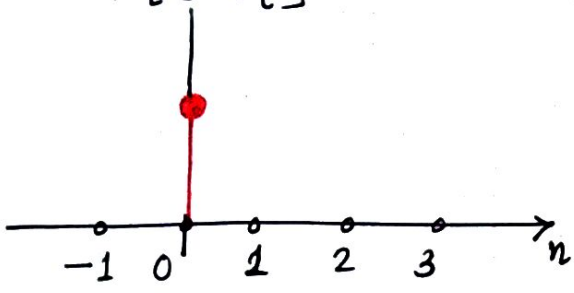
→ A transient input may lead to persitent output when there is feedback.

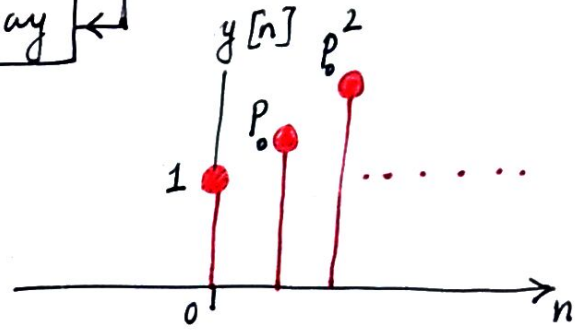


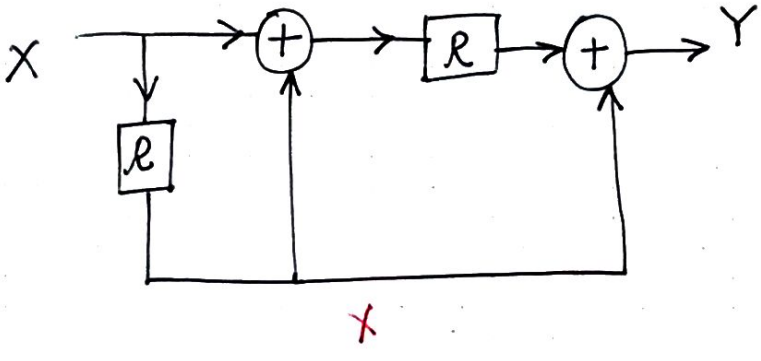


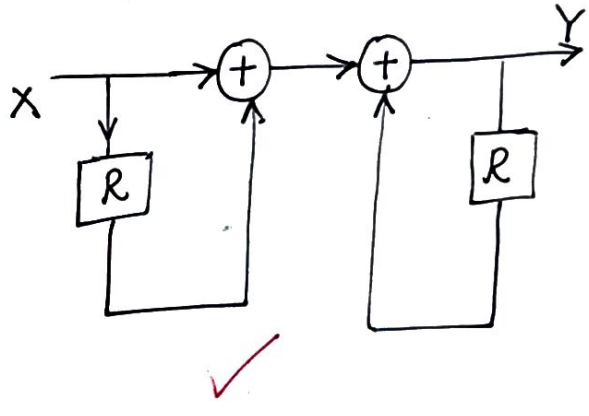
Feedback, Cyclic Signal Paths and Modes

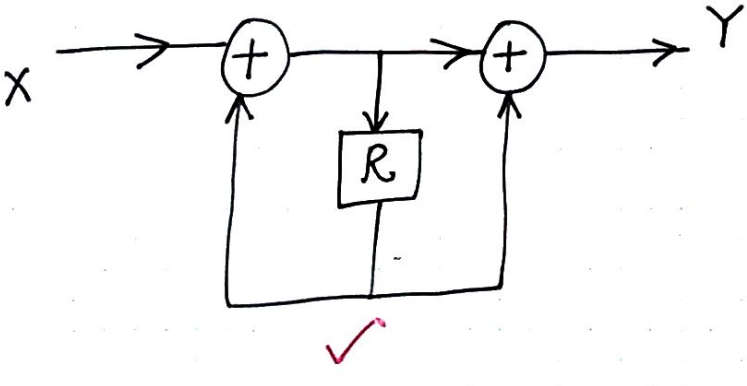


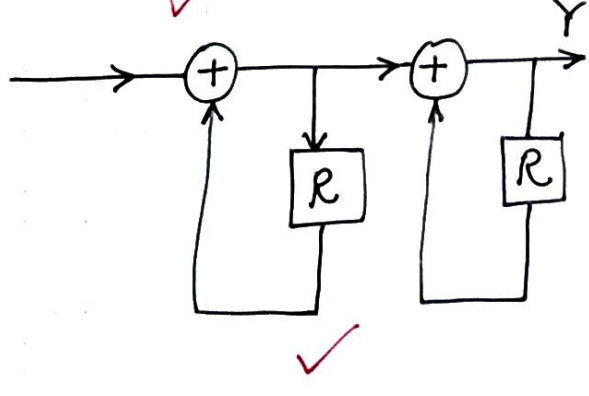




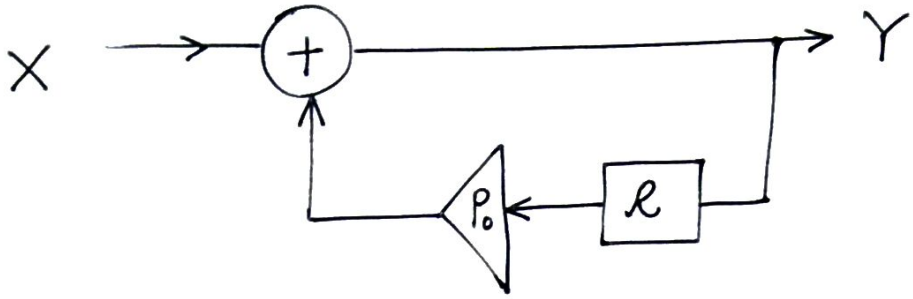




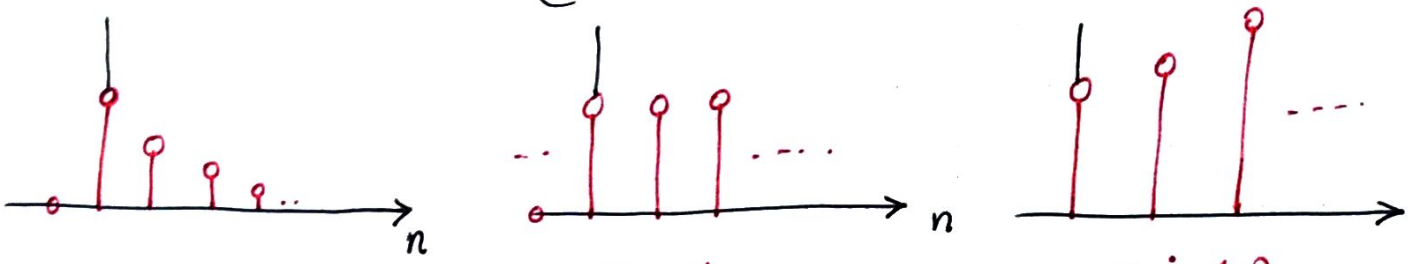




There is feedback if the output of some signal depends on its value at a previous time.



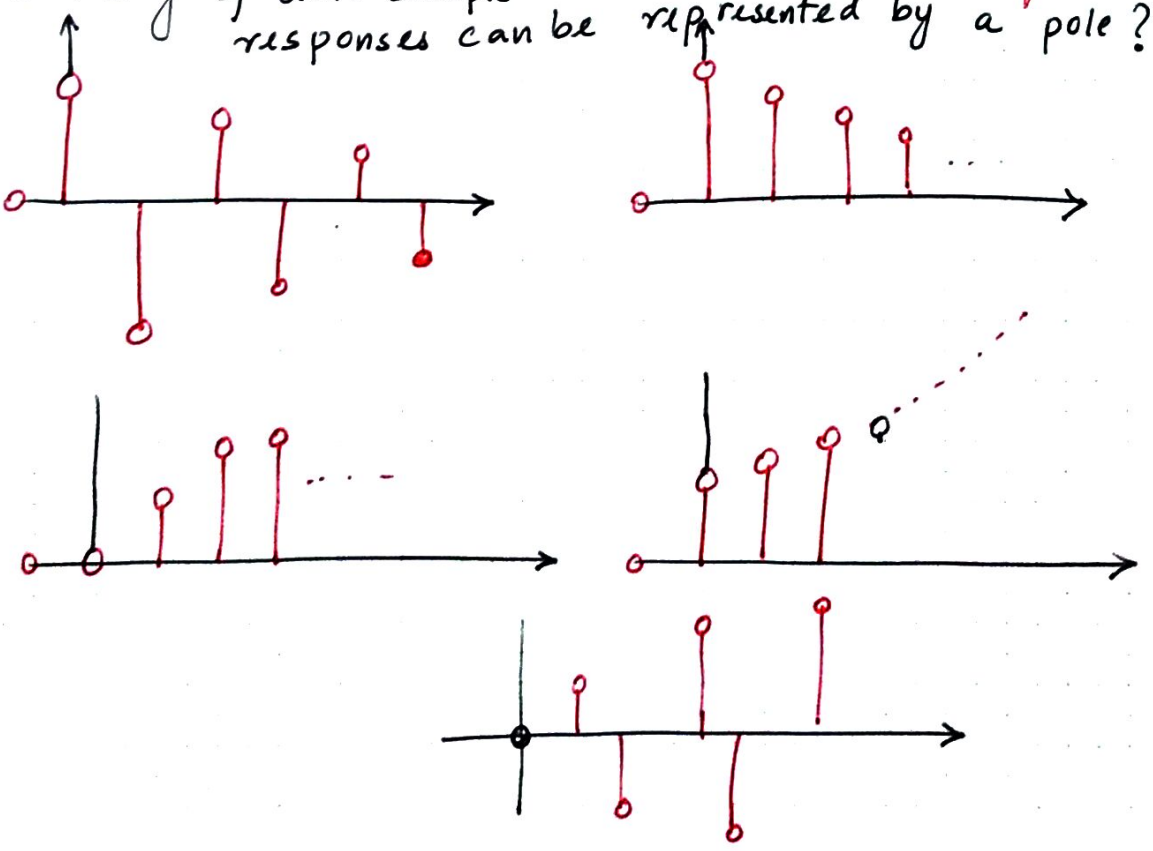
$$y[n] = \begin{cases} P_0^n, & \text{if } n > 0 \\ 0, & \text{otherwise} \end{cases}$$

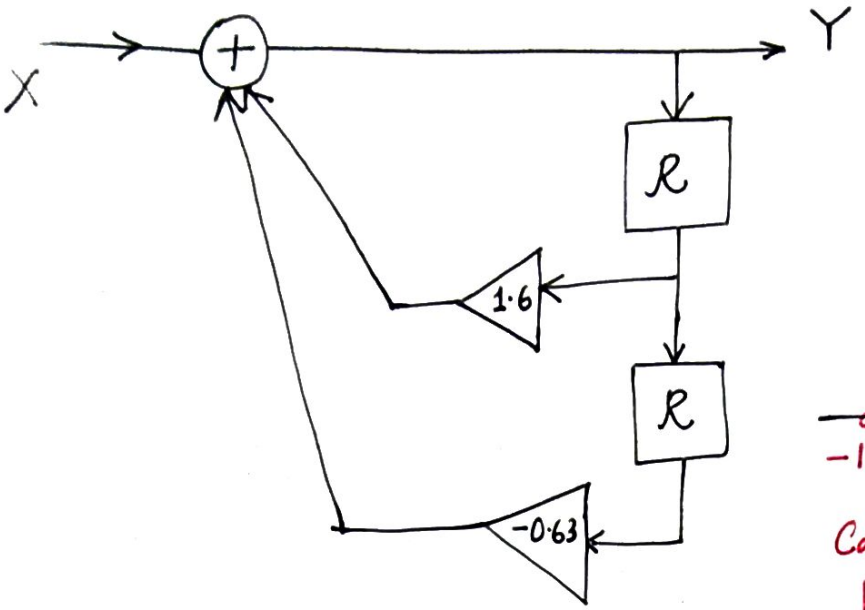


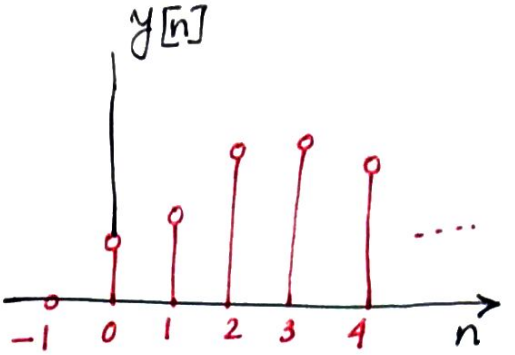
$P_0 = 0.5$ $\quad$ $P_0 = 1$ $\quad$ $P_0 \doteq 1.2$

P_0 is called pole. (Simple systems have this behavior characterized by a geometric sequence.)

How many of unit-sample responses can be represented by a pole?





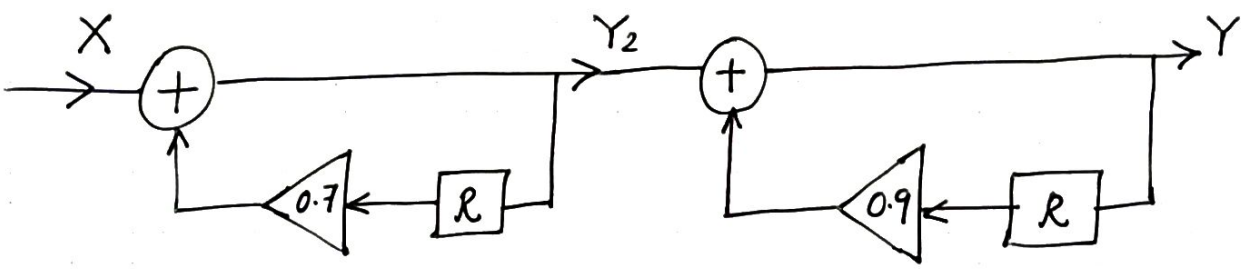


Can I represent it by a single pole?

$$Y = X + 1.6RY - 0.63R^2Y$$

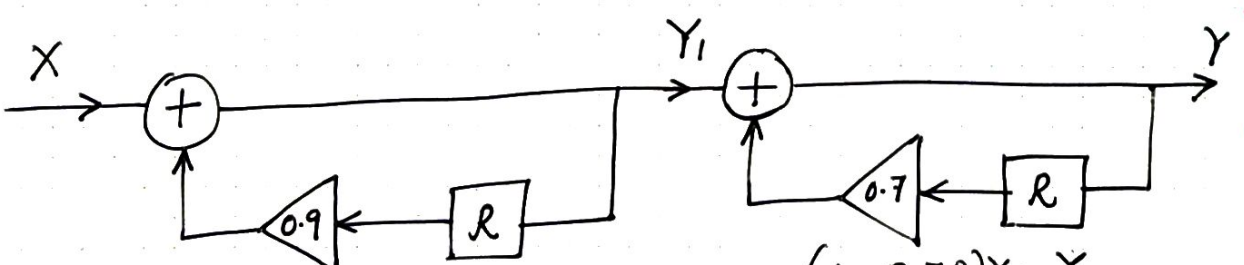
$$\Rightarrow Y - 1.6RY + 0.63R^2Y = X$$

$$\Rightarrow X = (1-0.7R)(1-0.9R)Y$$



$(1-0.7R)Y_2 = X$ $\qquad$ $(1-0.9R)Y = Y_2$

ORDER DOES NOT MATTER IF SYSTEM AT INITIALLY AT REST



$(1-0.9R)Y_1 = X$ $\qquad$ $(1-0.7R)Y = Y_1$

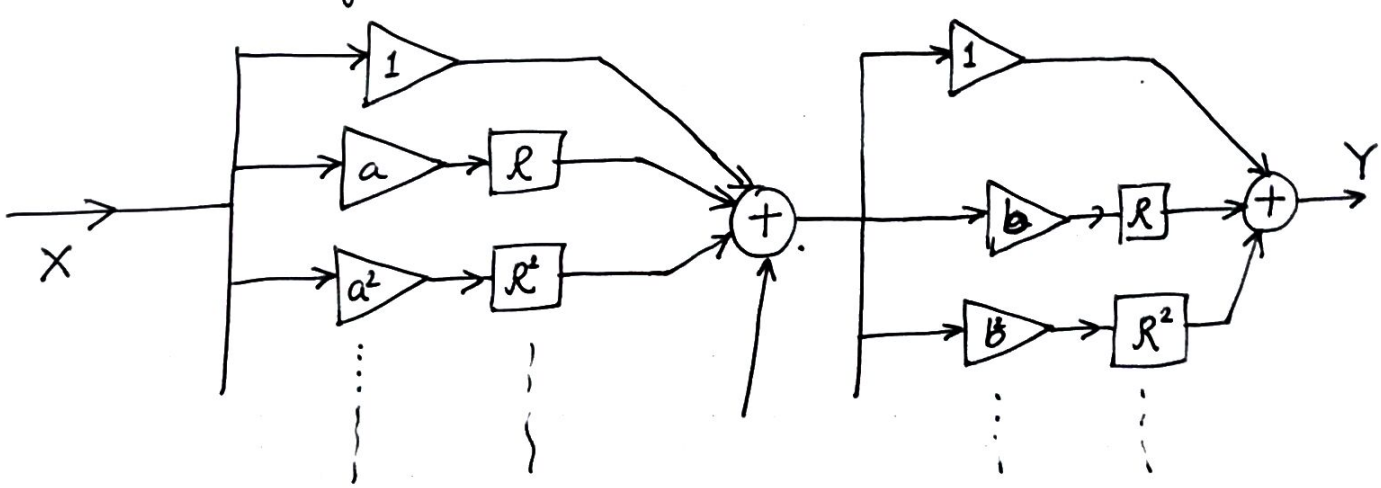
So, response of a complex system can be thought in terms of first order responses.

$$\frac{Y}{X} = \frac{1}{(1-0.7R)(1-0.9R)} = \frac{1}{(1-0.7R)} \times \frac{1}{(1-0.9R)}$$

$$\therefore Y = (1+0.7R+0.7^2R^2+\cdots) \times (1+0.9R+0.9^2R^2+\cdots)$$

$$= 1+(0.7+0.9)R+(0.7^2+0.9\times 0.7+0.9^2)R^2+\cdots$$

The only problem here is bookeeping. We can use use block diagrams:

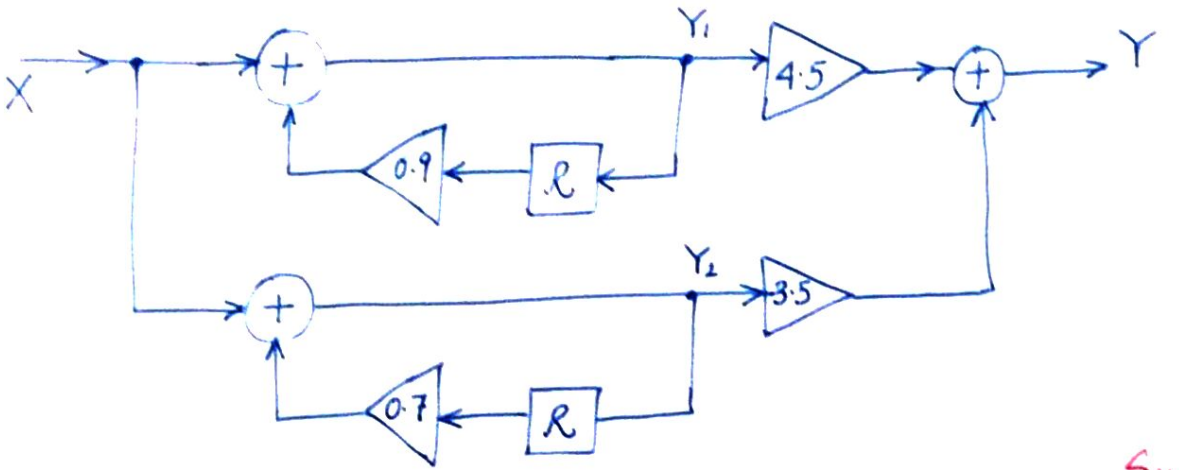


Collecting terms of equal order:

$$Y = 1+(a+b)R+(a^2+ab+b^2)R^2+\cdots\cdots$$

Partial fractions

$$\frac{Y}{X} = \frac{1}{(1-0.9R)(1-0.7R)} = \frac{4.5}{1-0.9R} - \frac{3.5}{1-0.7R}$$



$y_1[n] = 0.9^n u[n]$ and $y_2[n] = 0.7^n u[n]$

Sum of two geometric.

$$y[n] = \left[4.5\,(0.9)^n - 3.5\,(0.7)^n\right] u[n]$$

→ Partial fraction is a non-intuitive. Hence, polynomial representation of a system are so powerful.

$$Y = \frac{1}{(1-0.7R)(1-0.9R)} X$$

one pole another pole

Substitute $R \to 1/z$

$$Y = \frac{z^2}{(z-0.7)(z-0.9)}$$

roots of this polynomial are the poles.

e.g. $\boxed{y[n] = -\tfrac{1}{4}\, y[n-1] + \tfrac{1}{8}\, y[n-2] + x[n-1] - \tfrac{1}{2}\, x[n-2]}$

$$\left(1 + \tfrac{1}{4}R - \tfrac{1}{8}R^2\right) Y = \left(1 - \tfrac{1}{2}R\right) R\, X$$

$$\therefore \quad Y = \frac{R - \tfrac{1}{2}R^2}{1 + \tfrac{1}{4}R - \tfrac{1}{8}R^2}\, X$$

Substituting, $R = 1/z$.

$$Y = \frac{z - \tfrac{1}{2}}{z^2 + \tfrac{1}{4}z - \tfrac{1}{8}}\, X$$

$$= \frac{z - \tfrac{1}{2}}{\left(z + \tfrac{1}{2}\right)\left(z - \tfrac{1}{4}\right)}$$

Both poles inside unit circle so converges.

What are the poles in a Fibonacci system?

$$y[n] = x[n] + y[n-1] + y[n-2]$$

$$H = \frac{Y}{X} = \frac{1}{1 - R - R^2}$$

$$H(z) = \frac{z^2}{z^2 - z - 1}, \quad z = \frac{1 \pm \sqrt{5}}{2} = \phi \text{ or } -1/\phi$$

golden ratio

$\therefore\ z = 1.618$ and -0.618

diverging alternating.

$$H = \frac{\phi/\sqrt{5}}{1-\phi R} + \frac{1/\phi\sqrt{5}}{1+1/\phi\, R}$$

$$\therefore\ h[n] = \underbrace{\frac{\phi}{\sqrt{5}}\phi^n u[n] + \frac{1}{\phi\sqrt{5}}(-\phi)^{-n} u[n]}_{1,\ 1,\ 2,\ 3,\ 5,\ 8,\ 13,\ \ldots\ldots}$$

$r e^{\pm j\Omega}$

$r^n e^{\pm j\Omega n}$

up and down

spiral in/out

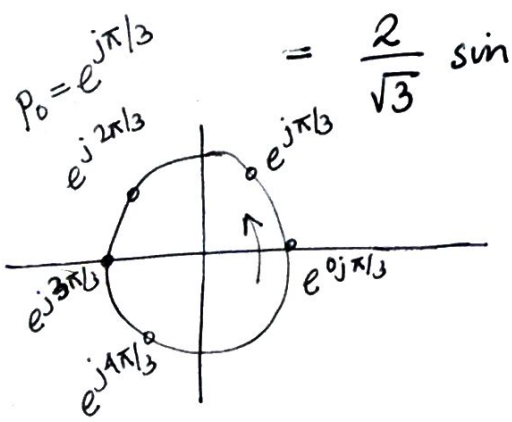
e.g. $y[n] = x[n] + y[n-1] - y[n-2]$

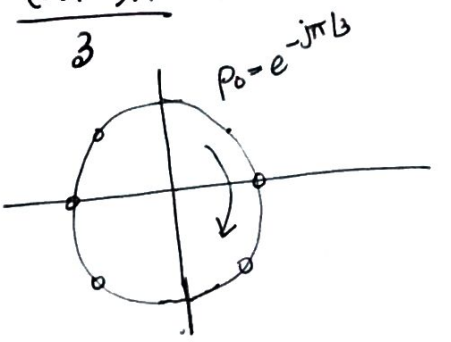
$$H = Y/X = 1/\left(1 - R + R^2\right)$$

$$= \frac{1}{1-e^{j\pi/3}R} \times \frac{1}{1-e^{-j\pi/3}R}$$

$$= \frac{1}{j\sqrt{3}}\left(\frac{e^{j\pi/3}}{1-e^{j\pi/3}R} - \frac{e^{-j\pi/3}}{1-e^{-j\pi/3}R}\right)$$

$$= \frac{2}{\sqrt{3}}\sin\frac{(n+1)\pi}{3}$$





Wierd that a real value system has complex poles.

But the output is always real.

Different ways of looking into systems —

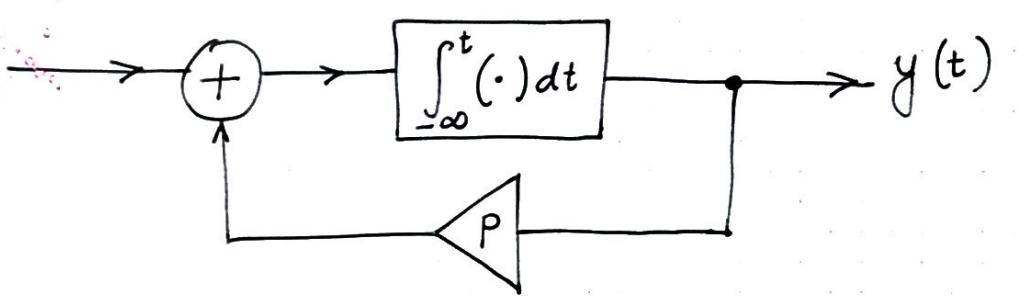
1. Difference equation: Concise and precise.
2. Block diagrams: Visualize the signal flow path.
3. Operator: Lets you treat systems like polynomials.

$\Updownarrow$ Same idea different result.

1. Differential equation: Mathematically compact.

$$\frac{d}{dt} y(t) = x(t) + p\,y(t)$$

2. Block diagrams: Illustrates signal flow path



3. Operator representations: Analyze systems as polynomials.

$$(1 - pA)Y = AX.$$

In DT the blocks were primarily made of delays. But since in CT the main operation is differentiation the blocks will be made of integrator.

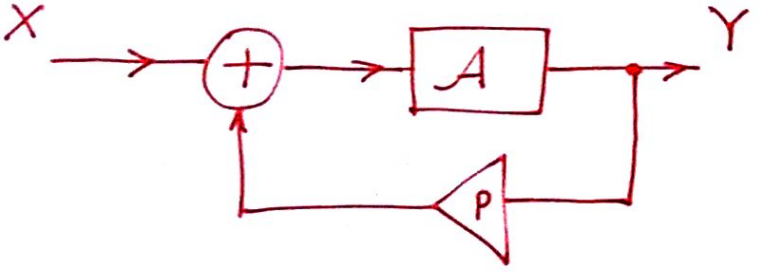
CT block diagrams are conciosly represented with the $\mathcal{A}$ operator.

Applying $\mathcal{A}$ to a CT signal generates a new signal that is equal to the integral of the first signal at all points in time.

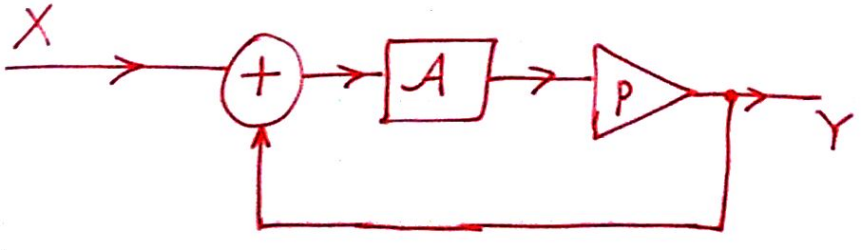
$$Y = \mathcal{A} X$$

is equivalent to

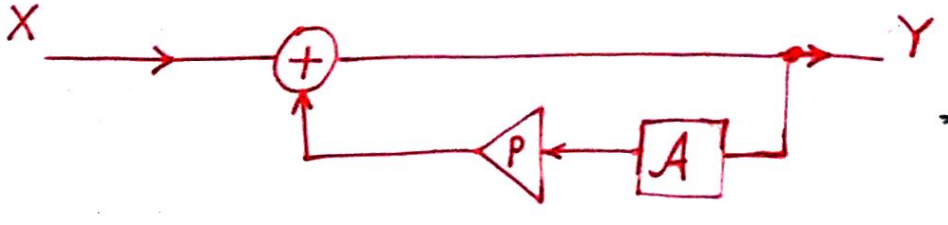
$$y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau \quad \forall\, t$$



$$\dot{y}(t) = x(t) + p\,y(t)$$



$$\dot{y}(t) = p\,x(t) + p\,y(t)$$



$$y(t) = x(t) + p\int y(t)$$

$$\rightarrow \dot{y}(t) = \dot{x}(t) + p\,y(t)$$